



QP CODE: 25801830

Reg No :

Name :

M.C.A DEGREE EXAMINATION, APRIL 2025

Second Semester

MASTER OF COMPUTER APPLICATION

CORE - MCACT202 - DATA STRUCTURES AND ALGORITHM ANALYSIS

2020 Admission Onwards

4CD01E62

Time: 3 Hours

Maximum: 75 Marks

Part A

*Answer any **ten** questions*

*Each question carries **3** marks*

1. Describe advantages of data structures.
2. What are practical complexities? How the practical complexities are being evaluated?
3. Convert the infix $(a+b)*(c+d)/f$ into postfix & prefix expression.
4. Write an algorithm to insert a node at the beginning of a linked list.
5. Define a tree.
6. Define a digraph.
7. What is binary search?
8. Define separate chaining method.
9. Represent a binary decision tree for Binary search with $n=8$.
10. Explain how greedy method is used in solving the knapsack problem.
11. Describe the control abstraction for backtracking.
12. What are the uses of comparison trees?

(10×3=30 marks)





Part B

Answer *all* questions

Each question carries **9** marks

13. a) Explain different operations performed on Queues with suitable algorithm and example.

OR

- b) Explain Time complexity evaluation of an Algorithm with suitable examples.

14. a) Explain search operation in binary search trees(BST).

OR

- b) Explain BFS with exmple and algorithm.

15. a) Explain the procedure of performing heap sort.

OR

- b) Why hash functions are needed? Explain any 3 hash functions.

16. a) Explain the Merge Sort algorithm with example.

OR

- b) Consider the instance of the knapsack problem $n=7$, $m=15$, $(p_1,p_2,\dots p_7)=(10,5,15,7,6,18,3)$ and $(w_1,w_2,\dots w_7)=(2,3,5,7,1,4,1)$. Find an optimal solution.

17. a) Explain the implementation of LC search with an example.

OR

- b) Explain any one algorithm for solving the multi-stage graph problem.

(5×9=45 marks)

